

ECE 647: Lecture 7

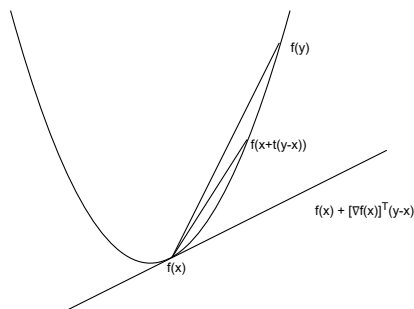
Sketch: Let us focus on the one-dimensional case: $f : \mathbb{R} \rightarrow \mathbb{R}$.

$\Rightarrow$ direction: If f is convex, then $(*)$ holds.

To see this, let

$$g(t) = \frac{f(x + t(y - x)) - f(x)}{t(y - x)}, \quad 0 < t \leq 1.$$

Note that $g(t)$ can be viewed as an approximation of the derivative $f'(x)$.



If $t = 1$, the ratio is

$$g(1) = \frac{f(y) - f(x)}{y - x}.$$

If $t \rightarrow 0$,

$$g(t) \rightarrow f'(x).$$

Graphically, it is intuitive that, as $t \rightarrow 0$,

$$g(t) \rightarrow f'(x),$$

which can be shown through convexity.

For us, in order to show that

$$\frac{f(y) - f(x)}{y - x} \geq f'(x),$$

it suffices to show that

$$g(t) \leq g(1) \quad \text{for all } t.$$

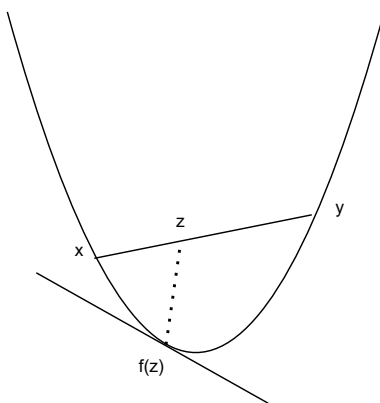
i.e.,

$$\frac{f(x + t(y - x)) - f(x)}{t(y - x)} \leq \frac{f(y) - f(x)}{y - x}.$$

" $\Leftarrow$ " direction: If (*) then f is convex.

For any x, y , let

$$z = \theta x + (1 - \theta)y.$$



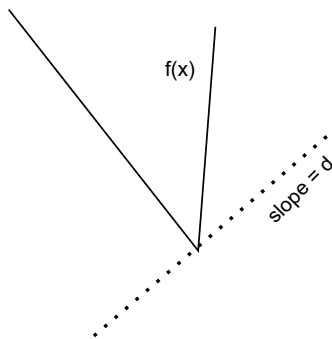
Q: What if the function is not differentiable?

- From the proof in the " $\Leftarrow$ " direction, the only property that we need is that there exists some vector d for every x .

$$f(y) \geq f(x) + d^T(y - x) \quad \text{for all } y.$$

This holds even if f is not differentiable at x .

$\Rightarrow$ Subgradients



- The first-order condition is useful when the second-order derivative is not easy to derive.

- Example: in some of the proofs for duality.

Assume that f is twice-differentiable, i.e., its Hessian $\nabla^2 f$ exists for each point in $\text{dom } f$.

Then f is convex if and only if $\text{dom } f$ is convex and

$$\nabla^2 f \geq 0 \quad \text{for all } x.$$

(Positive semidefinite)

f is strictly convex if $\text{dom } f$ is convex and

$$\nabla^2 f > 0 \quad \text{for all } x.$$

(Not necessary)

This is usually the most convenient criterion.

For functions on $\mathbb{R}$, we need $f''(x) \geq 0$.

For functions on $\mathbb{R}^n$, reduce to the one-dimensional case:

$$\begin{aligned} g(t) &= f(x + tv) \quad \text{for a given vector } v. \\ g'(t) &= [\nabla f(x + tv)]^T v. \\ g''(t) &= v^T \nabla^2 f(x + tv) v. \end{aligned}$$

We need $g''(t) \geq 0$ for all v
 $\Leftrightarrow \nabla^2 f(x)$ to be positive semidefinite.

Sketch: Focus on the one-dimensional case $f : \mathbb{R} \rightarrow \mathbb{R}$.

” $\Rightarrow$ ” **direction:** If f is convex, then $f''(x) \geq 0$.

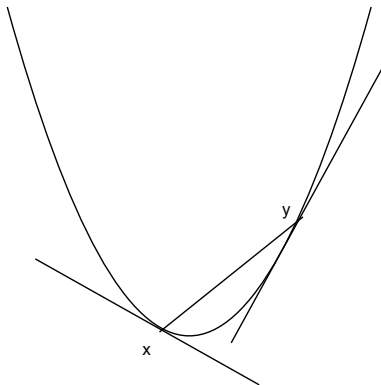
Using the first-order condition,

$$\begin{aligned} f(y) &\geq f(x) + f'(x)(y - x) \\ f(x) &\geq f(y) + f'(y)(x - y) \end{aligned}$$

$$\implies [f(x) - f(y)](x - y) \geq 0.$$

When $y > x$, we get

$$f'(y) \geq f'(x) \Rightarrow f''(x) \geq 0.$$



” $\Leftarrow$ ” **direction:** If $f''(x) \geq 0$, then f is convex.

Using the mean-value theorem,

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}f''(x + t(y - x))(y - x)^2$$

for some $t \in (0, 1)$. Since $f''(x + t(y - x)) \geq 0$, we obtain

$$f(y) \geq f(x) + f'(x)(y - x).$$

Thus, the result follows from the first-order condition.